Project report: Regularization in adjoint based inversion for frictional parameters in earthquake models

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Abstract

In earthquake rupture modeling, estimating frictional parameters from seismic observations is a highly nonlinear and ill-posed inverse problem. Small perturbations in the observed data can lead to large oscillations in the estimated parameters, making regularization necessary for stabilizing the inversion. In this study, we extend an existing adjoint-based inversion framework by introducing two regularization approaches: first-order Tikhonov (H1) regularization and total variation (TV) regularization, with the aim of stabilizing the inversion of the direct-effect parameter. The numerical experiments show that, for the baseline-resolution synthetic data, H1 regularization produces smoother and more stable estimates than TV. For the high-resolution case, however, H1 tends to oversmooth sharp discontinuities along the fault, whereas TV better preserves these features and provides a better balance between stability and resolution.

1 Introduction

In earthquake modeling, *rate–state friction laws* are commonly used to describe fault friction and involve several frictional parameters that strongly influence fault behavior. However, direct measurements of these frictional parameters are generally infeasible, and the parameters must therefore be inferred from seismic observations such as seismograms. This inference can be formulated as a PDE-constrained optimization problem, in which the optimal parameters are determined by minimizing the difference between model predictions and the corresponding observed data at specific receiver locations.

However, a brute-force finite-difference approach requires a separate model evaluation for each parameter perturbation in order to estimate the gradient, resulting in very high computational costs for high-dimensional inverse problems. [1] develops an adjoint-based inversion framework to infer frictional parameters from seismic observations. The adjoint method can compute the full gradient using only two model evaluations: one forward simulation and one adjoint simulation. This makes it considerably more efficient for high-dimensional inversions.

Despite this computational advantage, in real-world applications, the inferred frictional parameters can exhibit unphysical oscillations. These oscillations arise from the ill-posed nature of the inverse problem, where small perturbations in the observed data can lead to large variations in the inferred parameters. In this study, we use the publicly available code [2] associated with [1] and extend the existing adjoint-based inversion framework by introducing regularization terms. We evaluate two regularization approaches: first-order Tikhonov regularization (H1) and total variation regularization (TV).

The numerical experiments show that, for baseline-resolution synthetic data, H1 regularization produces smoother parameter estimates and more effectively suppresses oscillations than TV regularization. For the high-resolution case, however, H1 tends to oversmooth sharp discontinuities along the fault, whereas TV better preserves these features and provides a better balance between stability and resolution.

2 Forward model

The governing equations for a dynamic rupture model in two-dimensional antiplane shear are given by

$$\rho_\pm \ddot{u}_\pm = \partial_I \mu_\pm \partial_I u_\pm, \qquad \bar{x} \in \Omega_\pm, \quad t \in \mathcal{T}, \tag{2.1a}$$

$$u_\pm = u_{0\pm}, \quad \dot{u}_\pm = v_{0\pm}, \qquad \bar{x} \in \Omega_\pm, \quad t = 0, \tag{2.1b}$$

$$\tau_\pm + Z\dot{u}_\pm = 0, \qquad \bar{x} \in \partial\Omega_\pm \setminus \Gamma, \quad t \in \mathcal{T}, \tag{2.1c}$$

$$\tau_\pm = \mp F(V, \Psi), \qquad \bar{x} \in \Gamma, \quad t \in \mathcal{T}, \tag{2.1d}$$

$$\dot{\Psi} = G(V, \Psi), \qquad \bar{x} \in \Gamma, \quad t \in \mathcal{T}, \tag{2.1e}$$

$$\Psi = \Psi_0, \qquad \bar{x} \in \Gamma, \quad t = 0. \tag{2.1f}$$

Equation 2.1a represents the momentum balance for the displacement $u_\pm$, where the subscripts + and − denote the two sides of the fault Γ, as illustrated in Fig. 1. Equations 2.1b and 2.1c specify the initial and exterior boundary conditions, respectively.

In (2.1d), the slip velocity V is defined as the jump in particle velocity across the fault, $V = \dot{u}_+ - \dot{u}_-$, while Ψ denotes the state variable describing the contact strength of the fault interface. Together with the rate-and-state friction law (2.1d) and the state evolution equation (2.1e), Equations (2.1a)–(2.1f) form a coupled system that must be solved simultaneously for the displacement $u_\pm$ and the state variable Ψ.

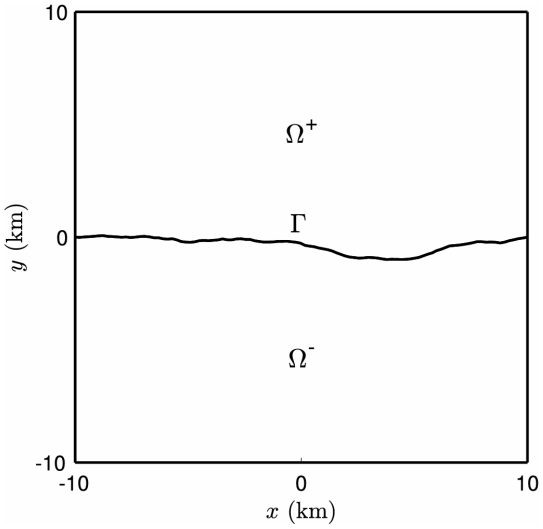


Figure 1: Domain with the fault Γ separating Ω_+ and Ω_-.
Reproduced from [1].

The rate-and-state friction coefficient $f(|V|, \Psi)$ is defined as:

$$f(|V|, \Psi) = a \sinh^{-1}\left(\frac{|V|}{2V_0} e^{\frac{\Psi}{a}}\right), \tag{2.2}$$

where a is the direct-effect parameter controlling the instantaneous response of friction to perturbations. As shown in Equation (2.2), the friction coefficient depends on the slip velocity $|V|$ and the state variable Ψ, while V_0 is an arbitrarily chosen reference slip velocity.

The rate-and-state friction law is obtained by equating the total shear stress to the total normal stress multiplied by the friction coefficient:

$$\tau_0 + \tau = \sigma_n^0 f(|V|, \Psi) \frac{V}{|V|} \tag{2.3}$$

We define the traction imbalance function as

$$F(V,\Psi) = \sigma_n^0 f(|V|,\Psi)\frac{V}{|V|} - \tau^0, \tag{2.4}$$

which represents the difference between the shear traction acting on the fault and the initial shear stress τ^0.

The evolution of the state variable is governed by the difference between the instantaneous friction coefficient $f(|V|,\Psi)$ and the steady-state value $f_{\rm ss}(|V|)$, as shown in Equation (2.5). If the slip velocity V remains constant in time, the friction coefficient gradually evolves toward its steady-state value, corresponding to steady sliding conditions on the fault.

$$G(V,\Psi) = -\frac{|V|}{D_c}\left(f(|V|,\Psi) - f_{ss}(|V|)\right), \tag{2.5}$$

The parameter a influences the fault dynamics by affecting both the instantaneous frictional response and the state evolution. Consequently, recovering a from observational data constitutes a highly nonlinear and ill-posed inverse problem. This ill-posedness makes the inversion highly sensitive to noise and can lead to high-frequency oscillations in the inferred parameters. We therefore introduce a regularization term to the objective functional to penalize large spatial variations in $a(\bar{x})$.

3 Misfit function

The objective of the optimization problem is to minimize a misfit functional subject to the governing PDE constraint. The misfit functional quantifies the difference between simulated and observed data at receiver locations, where the receivers are assumed to be seismometers recording ground motion. In essence, the goal is to determine the model parameters that minimize the residual between simulated and measured data at these locations.

At the k-th receiver, the residual is defined as

$$\bar{r}^{(k)}(t) = \bar{m}(\bar{x}_r^{(k)}, t) - \bar{m}_{data}^{(k)}(t). \tag{3.1}$$

where $\bar{m}(\bar{x}_r^{(k)}, t)$ denotes the model prediction at the k-th receiver and $\bar{m}_{\rm data}^{(k)}(t)$ is the corresponding observed signal. The total misfit functional is then obtained by summing the L^2-norms of the residuals over all $N_{\rm rec}$ receivers:

$$\mathcal{F} = \frac{1}{2}\sum_{k=1}^{N_{\rm rec}} \int_0^T \left|\bar{r}^{(k)}(t)\right|^2 dt. \tag{3.2}$$

4 Regularization and implementation

Regularization suppresses unrealistic oscillations in gradient-based optimization by adding penalty terms to the misfit functional. The total cost functional then reads.

$$\mathcal{F}_{\rm total} = \frac{1}{2}\sum_{k=1}^{N_{\rm rec}} \int_0^T \left|\bar{r}^{(k)}(t)\right|^2 dt + \mathcal{R}(p), \tag{4.1}$$

where $\mathcal{R}(p)$ is the regularization functional with a corresponding weighting coefficient λ that controls the smoothing strength. We consider three commonly used forms of regularization:

Tikhonov regularization:

Tikhonov regularization is a family of quadratic penalties that stabilize the inversion by constraining either the parameter amplitude or its spatial variation. Two commonly used forms are:

- **Zeroth–order Tikhonov:**

$$\mathcal{R}(p) = \lambda \int_\Gamma |p|^2 ds \approx \lambda P^T H_b P.$$

  Zeroth–order Tikhonov regularization applies an L^2 penalty directly on the parameter amplitude and does not involve spatial derivatives. It is the simplest and most commonly used variant, suitable for regularizing globally smooth parameter fields. It effectively suppresses noise, but does not explicitly enforce spatial smoothness.

- **First–order Tikhonov (H1):**

$$\mathcal{R}(p) = \lambda \int_\Gamma \left(\frac{dp}{ds}\right)^2 ds \approx \lambda (D_s p)^T H_b (D_s p).$$

  First–order Tikhonov regularization penalizes large spatial gradients of the parameters, thereby promoting smoothness along the fault. It efficiently suppresses high–frequency oscillations and is suitable for problems where the parameters vary smoothly and continuously, allowing gradual transitions but avoiding artificial oscillations.

Here, D_s represents the discrete derivative operator with respect to the local fault coordinate s, mapping the parameter values to their spatial gradients along Γ. And H_b denotes the diagonal integration weight matrix along the fault, providing the corresponding weights for integration.

Total Variation (TV) regularization:

$$\mathcal{R}(p) = \lambda \int_\Gamma \left|\frac{dp}{ds}\right| ds \approx \lambda\, 1^T H_b |D_s p|.$$

Total variation regularization penalizes the magnitude of spatial changes rather than their square. This allows local jumps or discontinuities without over-penalizing them. It is less sensitive to outliers and preserves sharp interfaces and discontinuities while suppressing spurious oscillations, making it particularly well suited for piecewise-constant or segmented fault properties.

Since the zeroth–order Tikhonov regularization does not involve spatial derivatives, the first–order (H1) Tikhonov regularization is adopted in this study, as sudden spatial changes occur along the fault. Together with TV, we test the effect of these two regularization terms in inversion experiment.

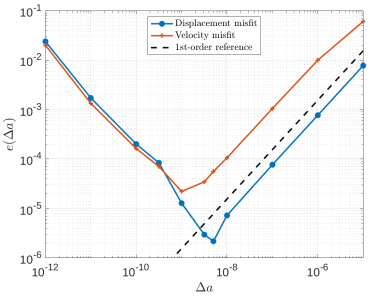
5 Computational results

5.1 Verification of the Regularization Term

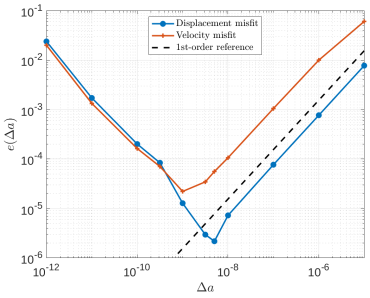
After adding the regularization term, we verify the correctness of the gradient computation. The computed error in Fig. 2 represents the difference between the adjoint gradient and the finite-difference gradient. The main purpose of this test is to demonstrate that the adjoint method produces the same gradient as the finite-difference approach, except for an error that originates entirely from the first-order finite-difference approximation.

When plotting this error as a function of the finite-difference step size Δa (or Δp), the curve initially decreases approximately proportionally to Δa, reflecting the dominance of truncation errors. As Δa becomes very small, round-off errors begin to dominate, and the error starts to increase approximately as $1/\Delta a$, as observed at around $\Delta a = 10^{-8}$. This characteristic behavior confirms that the gradient

evaluation is implemented correctly, thereby validating our approach and allowing us to proceed with the inversion.



(a) H1($\lambda = 10^{-3}$)



(b) TV($\lambda = 10^{-3}$)

Figure 2: Gradient Convergence.

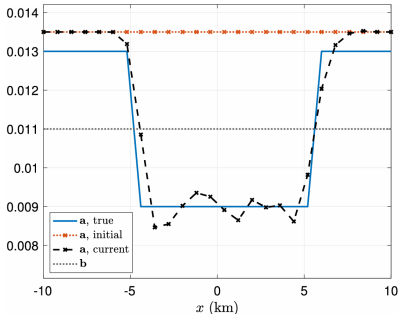
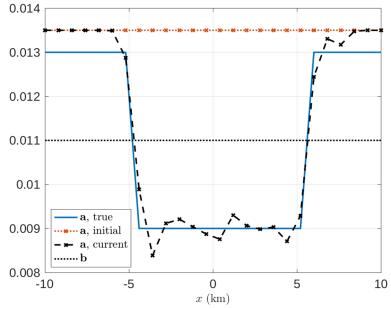


Figure 3: Reference Inversion ($m = 251, m_p = 26$)

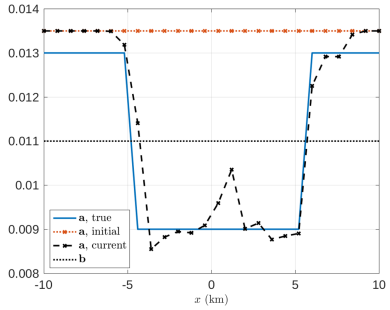
5.2 Inverse Crime

We now proceed to perform inversions in an inverse-crime setting, where the synthetic data are generated from a forward simulation under same physical settings as those used in the inversion. This experiment serves both to validate the methodology and to illustrate the behavior of the adjoint-based gradient optimization under idealized conditions.

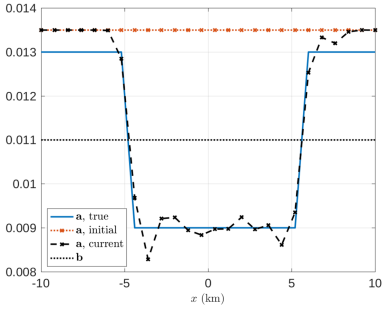
We generate synthetic data and infer the direct-effect parameter a using two types of regularization with different weights λ. This allows us to assess how regularization terms behaves before moving on to more realistic high-resolution data.The original unregularized inversion result from [1] is shown in Fig. 3, while the corresponding regularized inversion results are presented in Fig. 4 and Fig. 5.



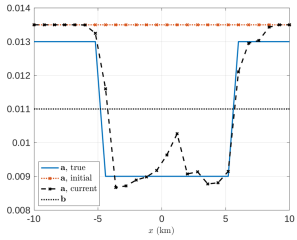
(a) $\lambda = 10$



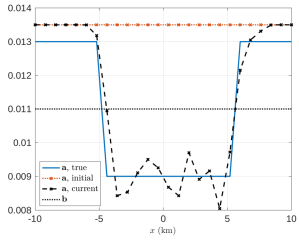
(b) $\lambda = 1$



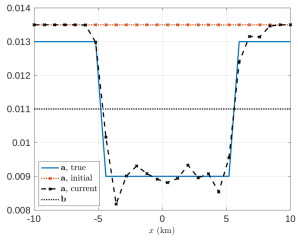
(c) $\lambda = 10^{-1}$



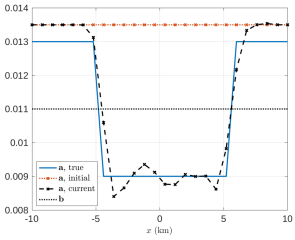
(d) $\lambda = 10^{-2}$



(e) $\lambda = 10^{-3}$

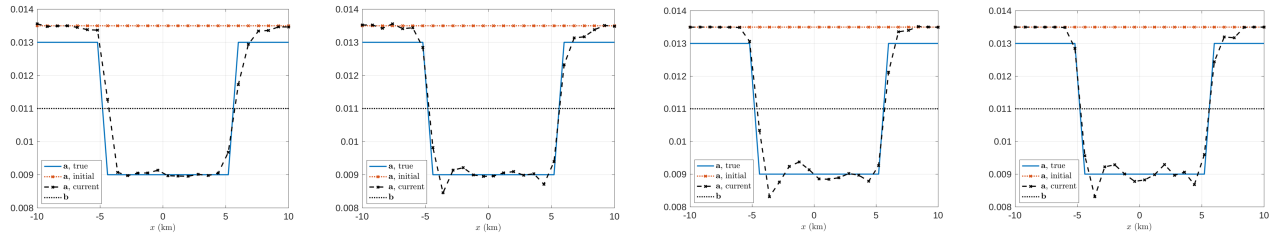


(f) $\lambda = 10^{-4}$

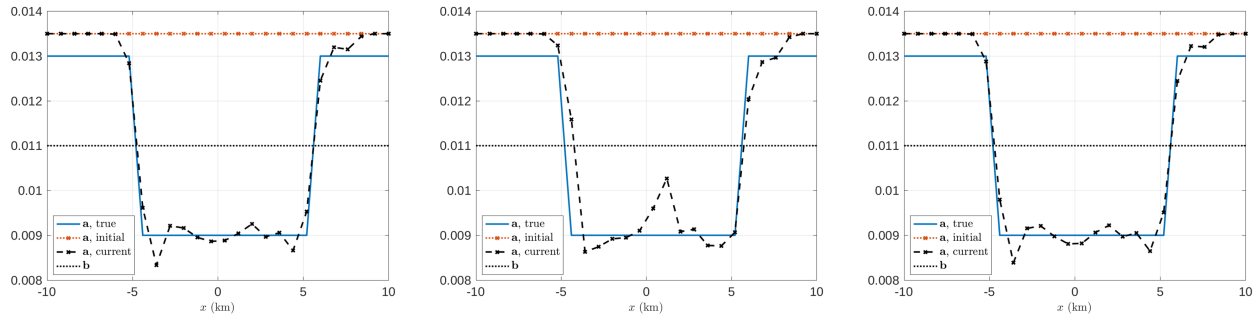


(g) $\lambda = 10^{-5}$

Figure 4: Inverted a with H1 ($m = 251$, $m_p = 26$).



(a) $\lambda = 10$ **(b)** $\lambda = 1$ **(c)** $\lambda = 10^{-1}$ **(d)** $\lambda = 10^{-2}$



(e) $\lambda = 10^{-3}$ **(f)** $\lambda = 10^{-4}$ **(g)** $\lambda = 10^{-5}$

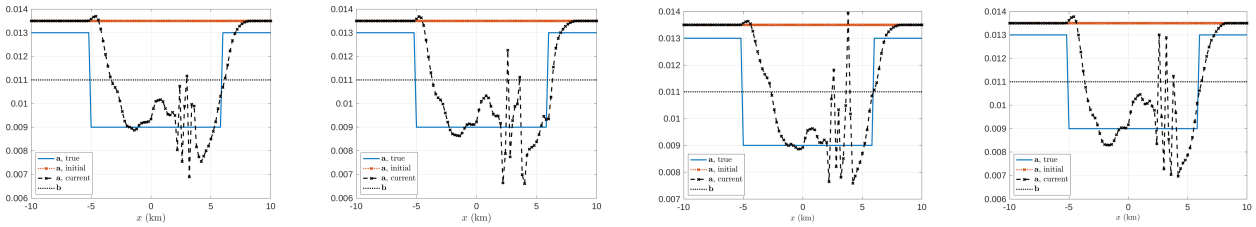
Figure 5: Inverted a with TV ($m = 251$, $m_p = 26$).

5.2.1 Result with increased parameter resolution

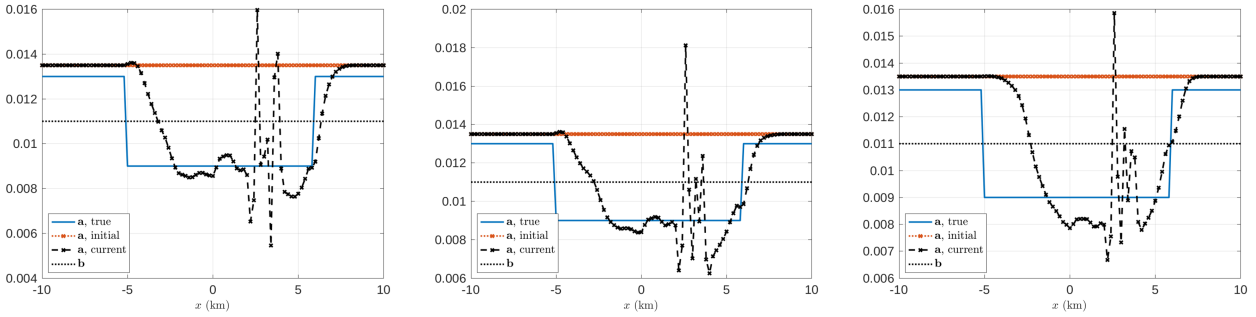
From Fig. 4 and Fig. 5, the optimal regularization weights are found to be $\lambda = 10^{-3}$ for TV regularization and $\lambda = 10$ for H^1 regularization, with the former showing a clear advantage in suppressing oscillations. However, under the current parameter resolution $m_p = 26$, the improvement remains unstable and not fully convincing.

Here, m denotes the number of grid points in the x-direction within the PDE solver, determining the spatial resolution of the forward simulation. A larger m provides finer spatial detail but also increases computational cost. Similarly, m_p denotes the number of discretized model parameters along the fault, controlling the resolution of the inverted parameters. A small m_p restricts spatial variability and produces smoother results, while a larger m_p allows for more detailed variations but also increases model complexity and the risk of high-frequency oscillations.

To better assess the effect of regularization, we increase m_p to 101, thereby introducing more degrees of freedom and higher potential for oscillatory behavior. To balance the increase in computational cost, we reduce the forward grid resolution by setting m to 126. This adjustment maintains computational feasibility while having negligible influence on the overall inversion quality. The results for both regularization terms are shown in Fig. 6 and Fig. 7.

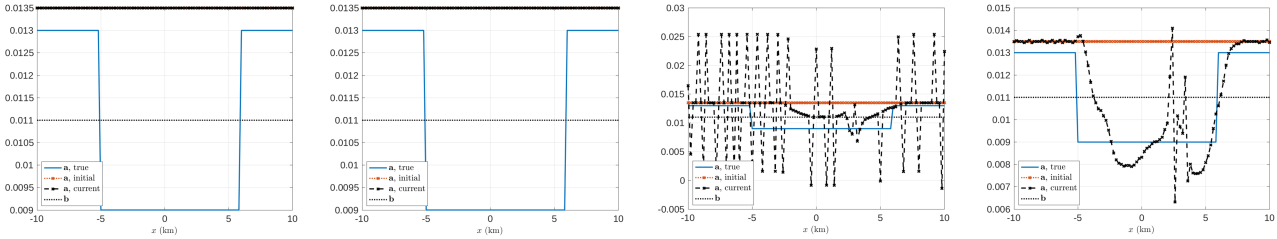


(a) $\lambda = 10$ **(b)** $\lambda = 1$ **(c)** $\lambda = 10^{-1}$ **(d)** $\lambda = 10^{-2}$

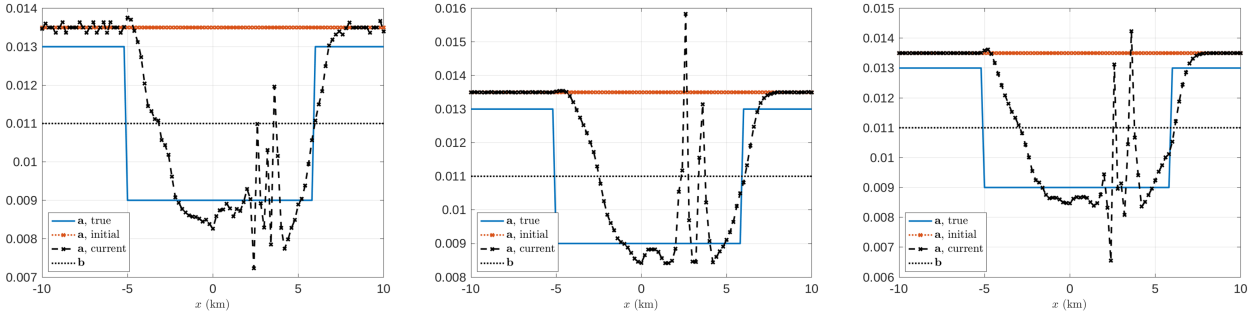


(e) $\lambda = 10^{-3}$ **(f)** $\lambda = 10^{-4}$ **(g)** $\lambda = 10^{-5}$

Figure 6: Inverted a with H1 ($m = 126$, $m_p = 101$).



(a) $\lambda = 10$ **(b)** $\lambda = 1$ **(c)** $\lambda = 10^{-1}$ **(d)** $\lambda = 10^{-2}$

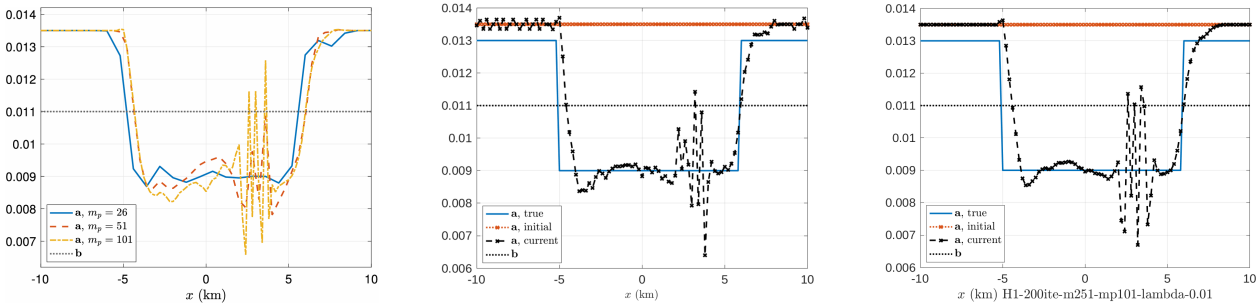


(e) $\lambda = 10^{-3}$ **(f)** $\lambda = 10^{-4}$ **(g)** $\lambda = 10^{-5}$

Figure 7: Inverted a with TV ($m = 126$, $m_p = 101$).

5.3 Result with increased iteration count

As we can observe from Fig. 6 to 7, it is confirmed that the optimal regularization weights for both regularization are found to be $\lambda = 10^{-3}$ for TV regularization and $\lambda = 10$ for H1 regularization. Using these optimal values, we increased the maximum number of iterations to 200 (with $m = 251$, $m_p = 101$) to observe the behavior of the inverse process, and compare it with the original behavior, as shown in Fig8a, yellow curve.



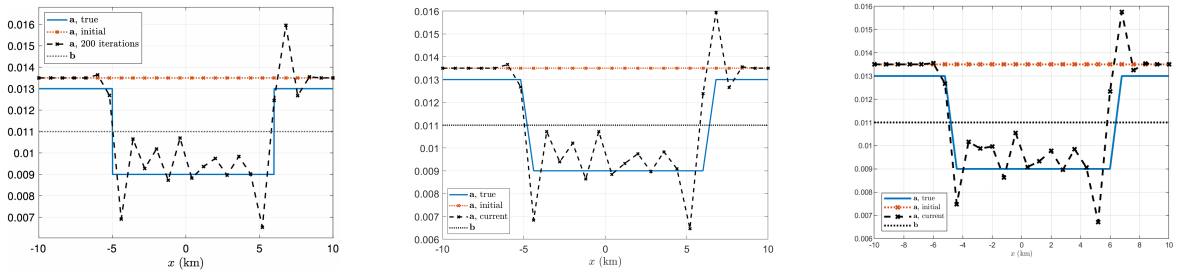
(a) Origin/Reference (200 iterations) **(b)** TV($\lambda = 0.001$) **(c)** H1($\lambda = 10$)

Figure 8: Comparison of inversions with 200 iterations ($m = 251$, $m_p = 101$).

We observe that the H1 regularization produces smoother inverted parameters near both boundary regions, whereas the TV regularization exhibits more small-scale oscillations at the same place. In the central part of the fault, however, the TV regularization performs slightly better in limiting the proportion of oscillatory areas, but it is less effective than H1 in suppressing the amplitude of these oscillations. Overall, neither regularization achieves the desired level of smoothness in the inversion results.

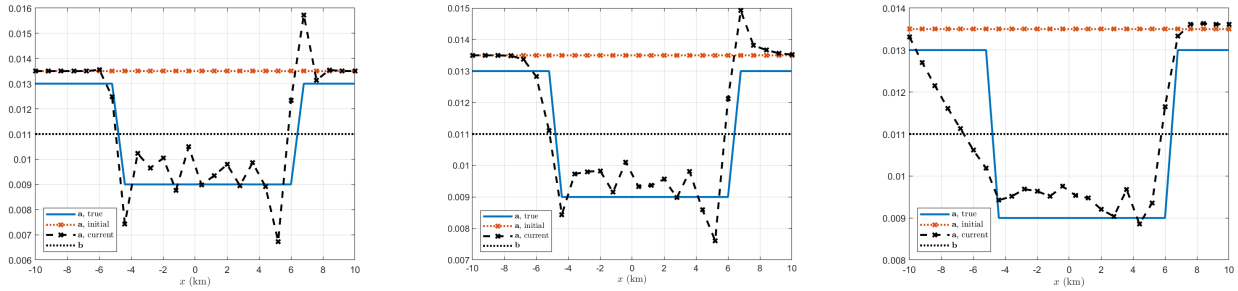
5.4 Inversion using high-resolution synthetic Data

We further evaluated the inversion framework using high-resolution synthetic data generated from a finer forward simulation. The point of the high-resolution data is that the forward model with the coarser grid cannot reproduce the data exactly.



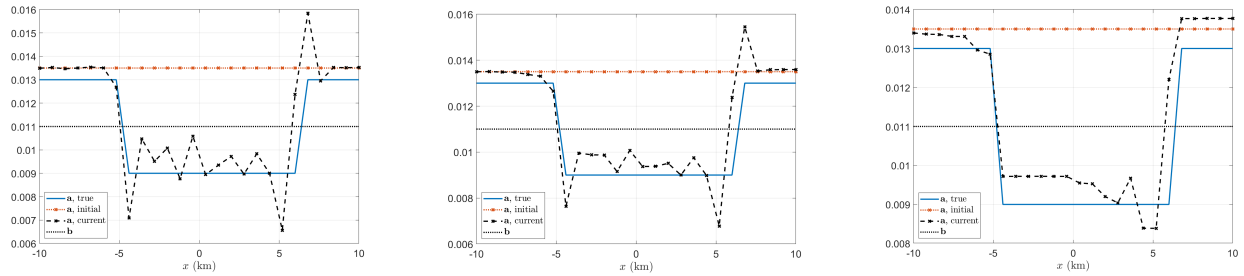
(a) High Resolution Data: Reference **(b)** TV, $\lambda = 10^{-3}$ **(c)** H1, $\lambda = 10$

Figure 9: High-resolution inversion using both regularization terms (200 iterations, $m = 251$, $m_p = 26$).



(a) H1, $\lambda = 100$ **(b)** H1, $\lambda = 1000$ **(c)** H1, $\lambda = 10000$

Figure 10: High-resolution inversion with H1 (200 iterations, $m = 251$, $m_p = 26$).



(a) TV, $\lambda = 10^{-1}$ **(b)** TV, $\lambda = 1$ **(c)** TV, $\lambda = 10$

Figure 11: High-resolution inversion with TV (200 iterations, $m = 251$, $m_p = 26$).

The results in Fig. 9 show that the H1 regularization with $\lambda = 10$ performs slightly better than the TV regularization with $\lambda = 0.003$. However, compared with the original result, the improvements introduced by both regularization terms remain minor. Consider similar results in normal synthetic data, we assume this is due to the relatively small magnitude of the regularization coefficient λ compared with the scale of the high-resolution data.

To further investigate, λ was gradually increased to 10 for TV regularization, and to 1000 for H1 regularization, to examine their effect on suppressing oscillations in the inversion results (as shown in Fig. 10 and Fig. 11). Interestingly, although H1 regularization shows a clear advantage over TV regularization in inversions with synthetic data, the difference becomes smaller for high-resolution data, particularly when λ is increased by several orders of magnitude. And it shows that H1 regularization has a stronger tendency to oversmooth sharp jumps as we keep increasing λ.

6 Conclusion and further discussion

When comparing the results with and without regularization, we observe that the regularization term does not significantly suppress oscillations with relatively small λ. However, an internal comparison with different values of λ reveals a slight improvement when λ increases. This suggests that the limited effect of regularization may be due to a mismatch between the scaling of λ and the misfit value.

Since the effect of conventional regularizations appears limited, in future research, we will explore two directions to improve the stabilization of the inversion by modifying the regularization strategy.

Scaling λ according to the misfit value

A straightforward approach is to test a range of λ values (e.g., 10^{-2} to 10^{5}) and assess their influence on the inversion outcome. However, such parameter sweeps can be computationally expensive.

To make the choice of λ more consistent and comparable across different cases, an alternative is to normalize the misfit function, for example, by dividing it by its mean value. This normalization ensures that the misfit and regularization terms are of comparable magnitude, allowing λ to be selected within a reasonable range. However, even after normalization, λ still needs to be tuned to achieve the desired balance between data fitting and model smoothness.

Multi-term Regularization

Another strategy is to combine multiple types of regularization terms to take advantage of their complementary properties. This approach combines different derivative-based penalties for example, H1 and TV within a single formulation. This approach allows the model to remain smooth in gradually varying regions while preserving sharp transitions where physical discontinuities occur. Meanwhile,

the relative weights of the individual terms also need to be chosen by scaling to the misfit value to ensure sufficient effect.

References

[1] Vidar Stiernström, Martin Almquist, and Eric M. Dunham. Adjoint-based inversion for stress and frictional parameters in earthquake modeling. *Journal of Computational Physics*, 519: 113447, 2024. ISSN 0021-9991. doi: https://doi.org/10.1016/j.jcp.2024.113447. URL `https://www.sciencedirect.com/science/article/pii/S0021999124006958`.

[2] Vidar Stiernström and Martin Almquist. Testing and verification of numerical methods in adjoint-based inversion for stress and frictional parameters in earthquake modeling, September 2024. URL `https://doi.org/10.5281/zenodo.13776965`.